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1. Identify the Stakeholders and User Needs

The main stakeholders of the hotel room reservation application are hotel management, hotel staff, customers, guests, the system administrator, and the development/DevOps team. Each stakeholder has different expectations and responsibilities.

Stakeholder	Interest / Need
Hotel management / owner	Efficient hotel operations, room occupancy, revenue, reliable system
Receptionists / hotel staff	Manage reservations, room availability, and customer bookings
Customers	Find rooms, make reservations, pay, and manage their bookings
Guests / visitors	Browse rooms and get information before deciding to book
System administrator	Maintain the system, users, security, and technical operation
Development / DevOps team	Develop, test, deploy, monitor, and maintain the application

The application has several user roles. A **User** can browse rooms and view information about availability and prices. A **Customer** has additional functionality, such as making and managing reservations and paying for a room. **Receptionists** use the management system to manage reservations and room availability. **Managers** have broader management permissions, including managing rooms, prices, and hotel information. The **System Administrator** manages technical and access-related aspects of the application.

Users / roles in the application

Application role	Main permissions
User	Browse rooms, view information, prices and availability
Customer	Browse + make/manage reservations and pay
Receptionist	Access management system, manage reservations and room availability
Manager	Manage rooms, prices, availability, hotel information and possibly reports
System Admin	Manage system users, permissions and technical settings

2. Define the Overall Application Requirements

Functional requirements

ID	Requirement
FR1	The application shall allow users to browse available hotel rooms.
FR2	The application shall display room information, including price, description, and availability.
FR3	The application shall allow customers to search for rooms based on selected dates.
FR4	The application shall allow customers to create a reservation.
FR5	The application shall allow customers to view and cancel their own reservations.
FR6	The application shall allow customers to make payments for reservations.
FR7	The application shall allow receptionists to create and manage customer reservations.
FR8	The application shall allow authorized staff to manage room availability.
FR9	The application shall allow managers to add, modify, and remove room information and prices.
FR10	The application shall prevent double booking of the same room for overlapping dates.
FR11	The application shall send a reservation confirmation to the customer after a successful booking.

Performance requirements

ID	Requirement
PR1	Normal user requests should receive a response within 2 seconds under normal load.
PR2	Room availability searches should return results within 2 seconds under normal load.
PR3	The system should support the expected number of simultaneous users without significant performance degradation.

Usability requirements

ID	Requirement
UR1	Users shall be able to browse rooms without creating an account.
UR2	The reservation process should be simple and understandable.
UR3	The interface should provide clear validation and error messages.
UR4	The application should be usable on desktop and mobile devices.

Reliability and availability

ID	Requirement
RR1	The application shall prevent data loss during normal operation.
RR2	Reservation data shall remain consistent if an operation fails.
RR3	The system shall automatically recover from temporary service failures where possible.
RR4	The application should be available continuously except during planned maintenance.
RR5	Database backups shall be created regularly.

Security

ID	Requirement
SR1	Users shall only access functionality permitted by their role.
SR2	Customers shall only be able to view and manage their own reservations.
SR3	Sensitive customer and payment information shall be protected.
SR4	Passwords shall not be stored as plain text.
SR5	Communication between the client and server shall use HTTPS.
SR6	Administrative functions shall require authentication and authorization.

Maintainability

ID	Requirement
MR1	The application shall use a modular architecture so individual components can be modified independently.
MR2	The application shall have automated tests for important functionality.
MR3	The application shall use version control.
MR4	The application shall use automated CI/CD pipelines for building and testing.
MR5	Application configuration shall be separated from source code.
MR6	The system shall provide logging to help diagnose errors and failures.

Compatibility and integration

ID	Requirement
CR1	The application shall provide a REST API for communication between the frontend and backend.
CR2	The application shall use a relational database for storing reservation and room data.
CR3	The application shall integrate with a payment service for customer payments.

ID	Requirement
CR4	The application shall be accessible through modern web browsers.
CR5	The application shall support deployment using containers.

Constraints

ID	Constraint
C1	The application shall comply with applicable data protection legislation, such as GDPR.
C2	The application shall use the technologies selected for the project, such as React, ASP.NET Core, PostgreSQL, and Docker.
C3	The project shall use Git for version control.
C4	The application shall be deployable using the project's CI/CD pipeline.
C5	The project shall be developed within the available course time and resources.

3. Partition the application into logical subsystems

1. User & Authentication Subsystem

Purpose: Manage users, accounts, authentication, and permissions.

Responsibilities:

- User registration/login
- Authentication
- Role management
- Authorization

Inputs:

- Login/registration data
- User role information

Outputs:

- Authentication result
- User information
- Access permissions

Functional requirements:

- The system shall allow users to log in.
- The system shall assign appropriate permissions based on the user's role.
- Customers shall only access their own reservations.

Non-functional requirements:

- Passwords must be securely stored.
- Authentication must use secure communication.
- Unauthorized users must not access protected functionality.

2. Room & Availability Subsystem

Purpose: Manage hotel rooms and their availability.

Responsibilities:

- Store room information
- Display room details
- Manage room availability
- Manage prices

Inputs:

- Room information
- Selected dates
- Availability changes from staff

Outputs:

- Available rooms
- Room details
- Prices
- Availability information

Functional requirements:

- Users shall be able to browse rooms.
- Customers shall be able to search availability by dates.
- Authorized staff shall be able to add, modify, and remove rooms.

Non-functional requirements:

- Availability searches should respond within 2 seconds under normal load.
- Room data must remain consistent.

3. Reservation Subsystem

Purpose: Create and manage hotel reservations.

Responsibilities:

- Create reservations
- View reservations
- Modify/cancel reservations
- Prevent double bookings

Inputs:

- Customer information
- Room
- Check-in/check-out dates
- Reservation changes

Outputs:

- Reservation confirmation

- Reservation details
- Updated room availability

Functional requirements:

- Customers shall be able to make reservations.
- Customers shall be able to view and cancel their reservations.
- Receptionists shall be able to manage customer reservations.
- The system shall prevent overlapping reservations for the same room.

Non-functional requirements:

- Reservation operations must be reliable.
- Failed transactions must not leave inconsistent booking data.

4. Payment Subsystem

Purpose: Handle payments for reservations.

Responsibilities:

- Process payments
- Associate payments with reservations
- Store payment status

Inputs:

- Reservation information
- Payment information

Outputs:

- Payment confirmation
- Payment status
- Failed-payment notification

Functional requirements:

- Customers shall be able to pay for their reservations.
- The system shall record payment status.
- The system shall associate a successful payment with the corresponding reservation.

Non-functional requirements:

- Payment information must be protected.
- The system should use a secure external payment provider rather than storing card details directly.

5. Administration & Management Subsystem

Purpose: Provide hotel staff and managers with management functionality.

Responsibilities:

- Manage rooms
- Manage prices

- Manage availability
- Manage reservations
- View management information

Inputs:

- Room changes
- Price changes
- Availability changes
- Reservation changes

Outputs:

- Updated room information
- Updated availability
- Updated reservation information
- Reports/statistics if implemented

Functional requirements:

- Receptionists shall be able to manage reservations.
- Managers shall be able to manage rooms and prices.
- Managers shall be able to manage room availability.

Non-functional requirements:

- Only authorized staff shall access the management system.
- Administrative actions should be logged.

4. Quality Function Deployment (QFD)

Step 1 — Customer/User needs (WHATs)

ID	Customer/User need	Importance
N1	Easily find available rooms	5
N2	Easily understand room information and prices	4
N3	Make a reservation easily	5
N4	Manage or cancel my reservation	4
N5	Make a secure payment	5
N6	Avoid double bookings	5
N7	Staff can efficiently manage reservations	5
N8	Staff can manage rooms and availability	4
N9	System is reliable and available	5
N10	Customer data is secure	5

Step 2 — Technical requirements (HOWs)

ID	Technical requirement
T1	Room availability search
T2	Room information and pricing database
T3	Reservation management API
T4	Customer reservation management
T5	Secure payment integration
T6	Booking validation and database constraints
T7	Staff management interface
T8	Room and availability management
T9	Automated backups and recovery
T10	Authentication, authorization and HTTPS
T11	Automated testing and CI/CD
T12	Logging and monitoring

Relationship between WHAT and HOW

User need ↓ / Technical requirement →	T1	T2	T3	T5	T6	T7	T8	T9	T10
N1 Find available rooms	•	•			•		•		
N2 Room information/prices		•					•		
N3 Make reservation	•		•		•				
N4 Manage reservation			•	•	•				•
N5 Secure payment			•	•					•
N6 Avoid double booking	•		•		•		•		
N7 Staff manage reservations			•		•	•			•
N8 Staff manage rooms		•				•	•		
N9 Reliability			•		•			•	
N10 Data security				•					•

5. Draw use-case and sequence diagrams for the application.